

ECS service force deployment POC

Requirements

1. Multiple ECS services with Fargate tasks.
2. Redeploy services regularly to avoid force upgrade
3. Send notifications

Architecture

EventBridge -> StepFunction -> Call API to forceDeploy -> SNS to send notifications

Sample

1. Create ecs services **fargate_test_service** in cluster **fargate_test** with a few tasks

Amazon Elastic Container Service > Clusters > fargate_test > Services > fargate_test_service > Tasks

fargate_test_service Info

↻ Update service Del

< Health and metrics **Tasks** Logs Deployments Events Configuration and networking

Tasks (1/5)

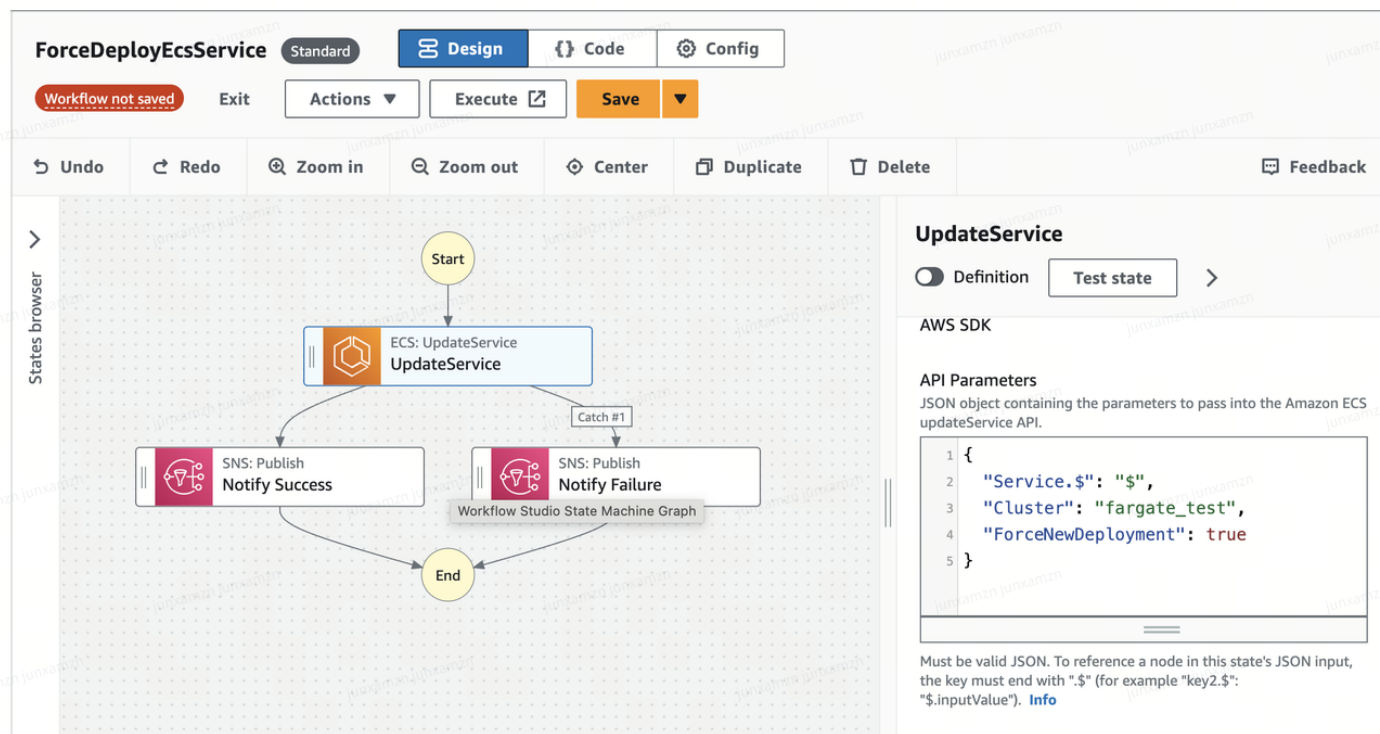
↻

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	Task ▾	Last status ▾	Desired st... ▾	Tas... ▾	Health status ▾
☒	397f9...	Running	Running	fargat...	Unknown
☐	4052f...	Running	Running	fargat...	Unknown
☐	4d818...	Running	Running	fargat...	Unknown
☐	6e44e...	Running	Running	fargat...	Unknown

2. Create Step Function “ForceDeployEcsService” to redeploy the service.

- “ECS: UpdateService” gets service name from inputPath. In our case, it will be defined in EventBridge Rule (See next step)
- SNS will publish success notification or catch failure.



Below are the codes for **UpdateService** FYI

```
"UpdateService": {
  "Type": "Task",
  "Next": "Notify Success",
  "Parameters": {
    "Service.$": "$",
    "Cluster": "fargate_test",
    "ForceNewDeployment": true
  },
}
```

3. Create event bridge rule to trigger step function

- In Step 2 “Define schedule” : Define the schedule pattern as needed
- In Step 3 “Select target(s) “: Select “Step function state machine” as target

c. You can Configure target input like below

```
{  
  "service_name": "fargate_test_service"  
}
```

4. Create SNS topic & Subscription to send notifications

Amazon SNS > Topics > Fargate_test_sns

Fargate_test_sns

EditDeletePublish message

Details

Name

Fargate_test_sns

ARN

arn:aws:sns:us-east-1:915286284889:Fargate_test_sns

Type

Standard

Display name

-

Topic owner

915286284889

Subscriptions

Access policy

Data protection policy

Delivery policy (HTTP/S)

Delivery status logging

Encryption

Tags

Integrations

Subscriptions (1)

EditDeleteRequest confirmationConfirm subscriptionCreate subscription

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ID

Endpoint

Status

Protocol

9d420d5d-2b29-497f-b8f3-0a2a604bbf58

junxamzn@amazon.com

Confirmed

EMAIL

Amazon SNS

Dashboard

Topics

Subscriptions

Mobile

Push notifications

Text messaging (SMS)

Origination numbers

New Feature

Amazon SNS now supports in-place message archiving and replay for FIFO topics. [Learn more](#)

Amazon SNS > Topics > Fargate_test_sns > Subscription: 9d420d5d-2b29-497f-b8f3-0a2a604bbf58

Subscription: 9d420d5d-2b29-497f-b8f3-0a2a604bbf58

EditDelete

Details

ARN

arn:aws:sns:us-east-1:915286284889:Fargate_test_sns:9d420d5d-2b29-497f-b8f3-0a2a604bbf58

Endpoint

junxamzn@amazon.com

Topic

Fargate_test_sns

Subscription Principal

arn:aws:iam::915286284889:user/jun

Status

Confirmed

Protocol

EMAIL

Subscription filter policy

Redrive policy (dead-letter queue)

Subscription filter policy

Info

This policy filters the messages that a subscriber receives.

No filter policy configured for this subscription.
To apply a filter policy, edit this subscription.

Edit